

WIP: Quality-Control Metrics for Crowdsourced Labeling: Promoting Dataset Reliability for Machine-Learning Applications

Aditya Iyer

*Department of Computer Science
North Carolina State University
Raleigh, NC, USA
asiyer2@ncsu.edu*

Tanmaiye Battula

*Department of Computer Science
North Carolina State University
Raleigh, NC, USA
tbattul@ncsu.edu*

Manoj Katravath

*Department of Financial Mathematics
North Carolina State University
Raleigh, NC, USA
mkatrav@ncsu.edu*

Edward F. Gehringer

*Department of Computer Science
North Carolina State University
Raleigh, NC, USA
efg@ncsu.edu*

Abstract—Crowd labeling has become a widely used approach for generating data required to train machine-learning models. However, the quality of the labels produced by crowdworkers varies significantly, leading to challenges in ensuring reliable datasets for model training. This study investigates strategies and metrics for assessing and improving crowd-sourced label quality. We identify markers of low-quality labeling and discuss their implications for dataset integrity and downstream machine-learning performance, emphasizing strategies for enhancing high-quality labeling.

Index Terms—Crowd labeling, quality control, crowdsourcing, dataset reliability, machine learning

I. INTRODUCTION

Machine learning relies on low-quality labeled data for its success. Creating labeled datasets using domain experts can be expensive and time-intensive, leading researchers to opt for crowdsourced labeling as a scalable and cost-effective alternative. By engaging a large number of non-expert contributors, crowd labeling enables the generation of diverse datasets for tasks such as image classification, sentiment analysis, and natural language processing [3, 4]. High-quality labeled data, however, remains the ultimate goal for ensuring optimal model performance.

Despite its advantages, crowd labeling introduces challenges related to the reliability, consistency, and engagement of contributors. For instance, some contributors may assign labels hastily or with insufficient understanding of the context, resulting in noisy or erroneous datasets [3, 4, 10]. These challenges have brought increasing attention to the need for robust quality control mechanisms to ensure the integrity and utility of crowd-labeled data.

II. NEED FOR QUALITY CONTROL

Crowd labeling involves distributing the task of data annotation to a large number of contributors, such as crowdworkers

or students, rather than relying solely on domain experts. This approach enables the creation of large and diverse datasets efficiently and affordably, which is critical for training machine-learning models in areas like natural language processing, image recognition, and sentiment analysis.

However, the effectiveness of crowd labeling hinges on the quality of the labels produced. Variations in engagement, understanding, and reliability among contributors can introduce significant noise into the data, [3, 4, 10], which, if left unchecked, can negatively impact the performance of machine-learning models. This highlights the importance of implementing stringent quality control measures to identify and address labeling errors.

A. Example Case Study

In this study, a web-based platform facilitating collaborative learning and peer assessment is used as a case study [8, 9] to explore crowd-labeling quality. Students are tasked with reviewing and critiquing the work of their peers, such as essays, project reports, or code repositories. To enhance the research process, students also categorize their comments using predefined tags, such as:

- **Suggestion:** The comment offers constructive suggestions for improvement.
- **Explanation:** The comment includes reasoning to justify feedback.
- **Example:** The comment provides examples to clarify points.

These categorizations serve as valuable training data for machine-learning models designed to assess the quality, constructiveness, and effectiveness of peer reviews.

III. RELATED WORK

Crowd labeling has emerged as a widely used technique for collecting annotated data, particularly for training machine learning models. However, ensuring the quality of this labeled data remains a significant challenge. While there is a rich body of literature on quality control in crowdsourcing, much of it is not directly applicable to our peer-assessment scenario. A substantial portion of prior work emphasizes privacy-preserving techniques, platform-level task routing, or worker incentivization, rather than focusing specifically on annotation reliability in educational contexts. Damon Runyon’s observation, “The race is not always to the swift, nor the battle to the strong; but that is the way to bet,” aptly encapsulates our approach: focusing our efforts on the subset of work that bets on label quality.

Some of the most relevant prior work includes methods for estimating inter-rater reliability (IRR), such as Krippendorff’s alpha, which balances observed and expected agreement among raters [8]. This has been applied to a variety of peer-assessed contexts, including MOOCs [9]. Our work builds on this foundation by introducing pattern and timing-based metrics to better capture engagement levels.

In the context of crowd labeling for ML, many studies have proposed fast-labeler filtering, based on the hypothesis that tags assigned in too little time are less reliable [3, 4]. However, concerns have been raised that such filtering may discard high-quality labels [11], and thus hybrid strategies have been proposed.

Pattern detection methods, particularly periodicity mining like the Periodicity Tree-based Verification (PTV) algorithm [7], are effective in detecting mechanical or inattentive labelers. Our use of binary tagging patterns and PR (Pattern Ratio) metrics extends this line of work.

Error-correcting codes (ECC) have also been proposed to handle noisy labels in crowdsourced settings [6], though much of the focus has been on privacy and secure label aggregation [5]. We adapt these ideas for educational peer feedback, using redundancy to infer consensus and measure reliability.

Several papers on crowdsourcing platforms [10, 13] explore broader operational and ethical aspects, including task assignment and worker incentives. While these provide useful background, they are not directly aligned with our goals of evaluating tagging quality and tagger credibility in peer learning environments.

In contrast to the general-purpose approaches in the literature, our work is rooted in a specific educational platform (Expertiza) and builds tagging-specific metrics tailored for peer review. By combining concepts from IRR, ECC, behavioral metrics, and pattern detection, our contribution lies in unifying these ideas into a coherent framework for filtering and calibrating student-generated labels.

IV. IDENTIFYING MARKERS OF LOW-QUALITY LABELING

Detecting low-quality or unreliable labels is essential to maintaining dataset integrity. Below are key behavioral mark-

ers that can indicate careless or inattentive labeling:

- **Speed of labeling** – Contributors who assign labels too quickly (e.g., faster than one tag per second) are unlikely to engage meaningfully with the task. Such rapid labeling increases the likelihood of hasty or arbitrary decisions, leading to labeling errors.
- **Repetitive tagging patterns** – Mechanical or highly predictable sequences such as “yes, yes, yes” or “yes, no, yes, no” suggest superficial engagement. These patterns reduce data credibility and can severely impact downstream machine learning performance.
- **Impact on dataset integrity** – The presence of noisy or inconsistent labels degrades dataset utility, compromising model accuracy and reliability. Identifying and addressing these low-quality behaviors is vital for ensuring robust training data.

V. METRICS FOR RELIABILITY

To promote the reliability of crowd-labeled data in educational peer assessment, we introduce four complementary metrics designed to detect low-quality tagging behavior. These metrics capture different dimensions of label reliability, including annotation speed, labeling patterns, alignment with peer consensus, and equivalence to expert validation. Together, they form a robust quality-control framework aimed at identifying inattentive or mechanical tagging, improving data integrity for downstream applications such as machine learning and educational analytics.

A. Too-Fast Tagging

One of the earliest indicators of low-quality annotations in peer assessment is excessively rapid tagging behavior. The core hypothesis behind the Too-Fast Tagging metric is that meaningful evaluation requires time; thus, responses entered too quickly are unlikely to reflect thoughtful judgment [3, 10]. This metric does not depend on the content of the tag itself but rather on the time intervals between successive tags assigned by a student. Tags submitted with unusually short intervals are flagged as potentially unreliable.

To account for natural variation and reduce the impact of outliers, we apply a logarithmic transformation to the raw tagging intervals. This compresses the range of time values and mitigates the influence of occasional long pauses (e.g., if a student is interrupted mid-task). Given a sequence of tagging times $\{t_1, t_2, \dots, t_n\}$, we compute the intervals as $\Delta t_i = t_i - t_{i-1}$, and apply the transformation:

$$\tilde{t}_i = \log(\Delta t_i + 1) \quad (5.1)$$

Here, $\tilde{t}_i$ is the log-transformed interval, and the addition of 1 ensures numerical stability even when Δt_i is very small.

To summarize a student’s overall tagging speed, we compute the geometric mean of these transformed intervals:

$$\bar{t}_{\text{geom}} = \exp \left(\frac{1}{n-1} \sum_{i=2}^n \log(\Delta t_i + 1) \right) \quad (5.2)$$

This value, $\bar{t}_{\text{geom}}$, reflects the average tagging interval on a compressed scale. Students whose average falls below a pre-determined threshold are flagged as tagging too fast, potentially indicating superficial engagement with the task [4], [10].

As an alternative approach to improve robustness, we also exclude the longest $x\%$ of intervals (e.g., top 5–10%), under the assumption that these represent interruptions or “coffee breaks” [3]. This trimmed set better reflects a student’s actual review pace by focusing on the intervals where active tagging likely occurred. The geometric mean is then calculated over the remaining intervals:

$$\bar{t}_{\text{trim}} = \exp \left(\frac{1}{|\mathcal{I}_{\text{trim}}|} \sum_{i \in \mathcal{I}_{\text{trim}}} \log(\Delta t_i + 1) \right) \quad (5.3)$$

Here, $\mathcal{I}_{\text{trim}}$ denotes the indices of the trimmed interval set. This method further stabilizes the tagging-time analysis and helps isolate patterns of inattentive behavior without penalizing occasional long breaks.

B. Pattern Detection

Pattern detection is used as a quality control strategy to identify reviewers who display repetitive or formulaic labeling behavior, such as marking every tag identically (e.g., “Yes, Yes, Yes”) or alternating between values in a predictable sequence (e.g., “Yes, No, Yes, No”) [3, 4]. Such regular patterns, especially when statistically improbable across diverse review content, may signal that the student is not engaging meaningfully with the review content but rather assigning labels arbitrarily or inattentively.

To detect such patterns, we analyze the binary tagging sequence assigned by each student across a review session. Each tag is converted into a binary format: “Yes” is encoded as 1, and “No” as 0, producing a sequence of values $\{x_1, x_2, \dots, x_n\}$, where $x_i \in \{0, 1\}$.

We define the Pattern Ratio (PR) as a high-level measure of labeling repetition:

$$\text{PR} = \frac{\text{Total Repeated Tags}}{\text{Total Tags Assigned}} \quad (5.4)$$

Since tagging responses are binary (e.g., “Yes” = 1 and “No” = 0), the Pattern Ratio effectively captures labeling uniformity. A tag is considered “repeated” if it matches the immediately previous tag. A high value of PR indicates a high concentration of repeated elements, such as consecutive identical tags or recurring sequences. For example, a PR of 0.8 means 80 percent of a student’s tags were part of a repeated segment. A threshold is set empirically; students exceeding this threshold are flagged as potentially unreliable [3].

To detect deeper structure in the sequence, we apply a periodicity detection method. A tag sequence $\{x_1, x_2, \dots, x_n\}$ exhibits periodic behavior of length p if:

$$x_i = x_{i+p} = x_{i+2p} = \dots \quad \text{for all valid } i \quad (5.5)$$

We search for such periods within a range $p \in [L_{\min}, L_{\max}]$, where $L_{\min}$ and $L_{\max}$ define the shortest and longest allowable

pattern lengths (e.g., 2 to 30). If such a periodic pattern is found, and the number of repetitions exceeds a minimum threshold R , the student’s tagging sequence is flagged for regularity [7].

In cases where the sequence includes noise (minor deviations), pattern detection can still identify approximate periodic structures using sliding windows and alignment checks [7]. A student’s pattern score increases with the number and strength of repeated substrings found.

This strategy is particularly effective for filtering out mechanical behaviors in labeling—such as clicking through tags without reading—thus enhancing the quality of the resulting labeled data for downstream applications like machine learning or rubric evaluation [3], [10].

C. Error-Correcting Codes for Tagging Reliability

In systems where multiple students independently review the same artifact, redundancy in review data allows us to apply techniques inspired by error-correcting codes (ECC) to detect inconsistencies and estimate the reliability of binary tagging behavior [6]. In our setup, each team submits an assignment and receives multiple peer reviews, which are then labeled by students as either helpful (1) or not helpful (0). These binary labels can be organized into a vector for each student’s tagging behavior.

Let the binary tagging vector assigned by a student over n reviews be $\mathbf{r} = [r_1, r_2, \dots, r_n]$, where $r_i \in \{0, 1\}$. Since these labels are inherently subjective, we do not assume access to a true ground truth. Instead, we construct a consensus vector based on majority agreement among all students who tagged the same reviews [1, 12].

For instance, suppose a particular review comment is labeled by 11 students, and 8 of them mark it as helpful. In this case, the consensus label is 1 (helpful), and the individual responses can be compared against this majority. To quantify disagreement, we use the Hamming distance, which measures the number of positions at which two binary vectors differ. Given two binary vectors $\mathbf{r}$ and $\mathbf{r}'$, the Hamming distance d_H is defined as:

$$d_H(\mathbf{r}, \mathbf{r}') = \sum_{i=1}^n \mathbb{1}(r_i \neq r'_i) \quad (5.6)$$

Let $\mathbf{m}$ be the consensus (majority vote) vector, constructed by taking the mode across all reviewers for each review. We consider a student’s tagging vector $\mathbf{r}$ to be reliable if it lies within a threshold θ of the consensus:

$$\text{Reliable if } d_H(\mathbf{r}, \mathbf{m}) \leq \theta \quad (5.7)$$

Vectors that exceed this Hamming distance threshold are flagged as potentially noisy or inattentive. When inconsistencies are identified, we apply a correction step via majority voting to estimate the most likely correct label:

$$\hat{r}_i = \text{mode}(r_i^{(1)}, r_i^{(2)}, \dots, r_i^{(k)}) \quad (5.8)$$

Here, $r_i^{(j)}$ represents the j th student’s response to the i th review. This strategy assumes the availability of at least $k \geq 3$ independent reviews to form a meaningful consensus [6, 13].

The effectiveness of this ECC-inspired strategy relies on the following data characteristics:

- **Redundancy:** Each submission must receive multiple independent reviews. In our dataset, this condition is satisfied as each comment typically receives 5 to 11 tags.
- **Consistency:** Students are instructed to follow a tagging rubric, reducing variance due to individual interpretation.
- **Distribution:** Reviews are randomly assigned to minimize systemic bias in tagging.

Although ECC traditionally assumes a fixed ground truth, our application adapts it to a crowdsourced setting by using the majority response as a *proxy for ground truth* [6]. This approach offers a practical way to evaluate reliability in a subjective peer assessment environment—making it especially suitable for our work-in-progress study, where absolute truth labels are not available but consensus-based inference is feasible.

D. Spot-Check Equivalence

Spot-Check Equivalence (SCE) is a reliability metric that quantifies the effectiveness of a crowd-labeling process by comparing it to an idealized scenario where a known percentage of data is manually spot-checked by experts. It was formally introduced by Burrell and Schoenebeck in the context of information elicitation [18], and has since been used in evaluating peer and model-based assessments without full ground truth [17].

SCE provides a scalable alternative to complete ground truth evaluation by measuring how well a peer-generated tagging or scoring mechanism replicates expert agreement using only a small set of validated labels. It also aligns with the findings of earlier crowd-versus-expert comparison studies [1], as well as partial expert supervision approaches seen in medical annotation contexts [14].

In the context of our peer-assessment platform, SCE estimates the proportion of expert spot-checked data required to match the reliability of a given scoring strategy (e.g., ECC, pattern detection). If a peer-labeling strategy achieves the same quality as an expert checking 30% of the data, its SCE is 0.3.

To compute SCE, we first define a performance proxy, such as the correlation between label reliability and tagging effort (e.g., number of comments tagged or time taken). We then determine the minimal expert-checking fraction that achieves the same proxy score. Mathematically, the SCE for a given strategy S is:

$$\text{SCE}(S) = \min_{x \in [0,1]} \{f(S) \geq f(\text{Expert}_x)\} \quad (5.9)$$

Here, $f(S)$ is a measurement function (e.g., Kendall’s tau or Spearman’s rank correlation) for strategy S , and Expert_x represents a spot-checking approach with x fraction of expert-validated labels.

Example: Assume that students are asked to tag review comments as to whether they were helpful or not. Suppose Student A’s labeling shows a 0.65 correlation with peer-verified helpfulness. If expert spot-checking of 25% of the reviews yields the same correlation (viz., 0.65), then $\text{SCE}(A) = 0.25$.

This metric is particularly useful when a full ground-truth determination is unavailable or infeasible. It enables practitioners to compare different label-validation strategies (e.g., ECC, peer prediction) against partial expert input, helping to quantify trade-offs between cost and accuracy.

Advantages:

- Provides an interpretable comparison to expert-checking baselines.
- Requires only partial ground truth, reducing labeling costs.
- Useful for evaluating metrics like ECC or tagging credibility.

Limitations:

- Requires some amount of expert-labeled data for calibration.
- Effectiveness depends on the robustness of the measurement proxy f .

This strategy aligns well with settings where expert resources are limited but peer-labeled data is abundant. It bridges the gap between fully supervised quality control and crowdsourced approximations.

VI. EDUCATIONAL IMPLICATIONS AND RESEARCH-TO-PRACTICE IMPACT

The proposed quality-control metrics not only improve the reliability of crowd-labeled datasets for machine learning applications but also hold significant promise for enhancing educational practice, particularly in peer assessment settings.

Promoting Better Learning: The proposed quality control metrics align closely with established educational theories. For instance, Constructivist learning theory emphasizes the importance of learners actively engaging with content and constructing meaning through social and cognitive processes. By identifying inattentive or mechanical tagging (e.g., via Too-Fast Tagging or Pattern Detection), instructors can foster deeper reflection and more authentic engagement in peer review, encouraging students to meaningfully interact with peer work rather than passively labeling it. Furthermore, Self-Regulated Learning (SRL) theory highlights metacognition, self-monitoring, and feedback loops. Metrics such as ECC and Spot-Check Equivalence provide a foundation for accountability and self-awareness, as students can be made aware of how their tagging aligns with peers or expert benchmarks. Additionally, Cognitive Load Theory (CLT) suggests that excessive or unclear demands on working memory reduce learning efficiency. If tagging instructions or task complexity lead to repetitive or too-fast tagging, these metrics can surface such cognitive overload signals, prompting a redesign of instructions to reduce extraneous load and enhance learning outcomes.

Formative Assessment Tool: Instructors can use these metrics to monitor the quality of peer assessments in real time. For instance, detecting rapid tagging or repetitive labeling patterns may indicate the need for clarification of instructions or additional scaffolding. This enables peer review activities to serve as ongoing formative assessment opportunities, rather than one-off grading tasks.

Grading Assistance and Calibration: Reliability indicators like ECC scores and Spot-Check Equivalence can also be used to adjust participation scores or provide partial automation in assigning credit for peer review contributions. By integrating these metrics into grading rubrics, educators can balance fairness with accountability, ensuring that thoughtful reviewers are appropriately recognized.

Scalability and Transferability: Because these metrics rely on lightweight behavioral data (e.g., tagging speed, sequence patterns), they can be applied at scale across large classrooms or online platforms. This opens pathways for integrating such metrics into MOOC platforms, flipped classrooms, or collaborative learning systems where traditional instructor-led feedback may not be feasible.

Collectively, these research-to-practice mechanisms offer a pathway to enhance the pedagogical effectiveness of peer learning while simultaneously supporting the integrity of machine-learning-ready educational data.

VII. FUTURE WORK

Building on the current study, future work will focus on incorporating expert-labeled data to better calibrate and validate our proposed reliability metrics, such as Error-Correcting Codes and Pattern Ratio. This integration will allow for a more precise computation of Spot-Check Equivalence and provide a stronger benchmark for evaluating crowd-labeled data quality.

Additionally, we plan to develop predictive models that utilize behavioral features like tagging speed, pattern repetition, and agreement with peers to estimate student engagement and label reliability in real time. Such models could enable dynamic quality control by identifying low-quality labeling as it occurs.

Another promising direction involves designing feedback mechanisms that inform students about the quality of their tagging, encouraging more thoughtful and reflective participation in peer assessment. By fostering awareness of their labeling behavior, students may improve the accuracy and usefulness of their contributions.

Longitudinal analysis will also be conducted to explore how labeling behaviors evolve over time or across multiple assignments. This could reveal important insights into learning progression, changes in engagement, or potential fatigue effects that impact data quality.

Finally, we aim to test the generalizability of our quality metrics by applying them to other crowdsourcing domains, such as image or text annotation tasks. This will help determine whether the proposed framework can be effectively transferred beyond educational peer review contexts to broader labeling applications.

VIII. CONCLUSION

Crowd labeling offers a scalable and cost-effective method for generating training datasets, but its success depends on rigorous quality control. This study highlights the need to detect low-quality contributions and enforce labeling standards to improve dataset integrity. By identifying markers of poor labeling and refining quality control processes, researchers can enhance the reliability of crowdsourced datasets and maximize downstream effectiveness in machine-learning applications.

REFERENCES

- [1] R. Snow, B. O'Connor, D. Jurafsky, and A. Y. Ng, "Cheap and fast—but is it good?: Evaluating non-expert annotations for natural language tasks," *EMNLP*, 2008, pp. 254–263.
- [2] T. Kulesza, M. Burnett, W. K. Wong, S. Stumpf, S. Perona, and A. J. Ko, "Principles of explanatory debugging to personalize interactive machine learning," *Proceedings of the 20th International Conference on Intelligent User Interfaces*, 2014, pp. 126–137.
- [3] F. Daniel, P. Kucherbaev, C. Cappiello, B. Benatallah, and M. Al-lahbakhsh, "Quality control in crowdsourcing: A survey of quality attributes, assessment techniques, and assurance actions," *ACM Comput. Surv.*, vol. 51, no. 1, pp. 7:1–7:40, 2019.
- [4] P. Upchurch et al., "Interactive consensus agreement games for labeling images," *HCOMP*, vol. 4, no. 1, pp. 239–248, 2016.
- [5] L. R. Varshney, "Privacy and reliability in crowdsourcing service delivery," *SRII Global Conf.*, 2012.
- [6] A. Vempaty, L. R. Varshney, and P. K. Varshney, "Reliable crowdsourcing for multi-class labeling using coding theory," *arXiv preprint arXiv:1309.3330*, 2013.
- [7] K.-Y. Huang and C.-H. Chang, "Mining periodic patterns in sequence data," *International Conference on Data Warehousing and Knowledge Discovery*, 2004.
- [8] K. Krippendorff, "Computing Krippendorff's alpha-reliability," *University of Pennsylvania ScholarlyCommons*, 2011. [Online]. Available: https://repository.upenn.edu/asc_papers/43
- [9] F. Garcia-Loro, S. Martin, J. A. Ruipérez-Valiente, E. Sancristobal, and M. Castro, "Reviewing and analyzing peer review Inter-Rater Reliability in a MOOC platform," *Comput. Educ.*, vol. 154, 2020.
- [10] M. Lease, "On quality control and machine learning in crowdsourcing," *AAAI Workshops*, 2011.
- [11] A. Kittur, J. V. Nickerson, M. Bernstein, E. Gerber, A. Shaw, J. Zimmerman, M. Lease, and J. Horton, "The future of crowd work," *CSCW*, 2013, pp. 1301–1318.
- [12] Y. Zhang and E. F. Gehringer, "Can students produce effective training data to improve formative feedback?," *Frontiers in Education* 2021.
- [13] D. R. Karger, S. Oh, and D. Shah, "Iterative learning for reliable crowdsourcing systems," *NeurIPS*, 2011, pp. 1953–1961.
- [14] E. Heim, "Large-scale medical image annotation with quality-controlled crowdsourcing," *Ph.D. dissertation, University of Heidelberg*, 2018.
- [15] G. Kazai and I. Zitouni, "Quality management in crowdsourcing using gold judges behavior," *WSDM '16: Proceedings of the Ninth ACM International Conference on Web Search and Data Mining*, 2016.
- [16] A. Som, A. Samanta, H. Shaikh, A. Mahmud, and A. S. Nath, "Automated student group collaboration assessment and recommendation system using individual role and behavioral cues," *Front. Comput. Sci.*, vol. 3, p. 728801, 2021.
- [17] S. Xu, Y. Lu, G. Schoenebeck, and Y. Kong, "Benchmarking LLMs' judgments with no gold standard," *arXiv preprint arXiv:2402.13567*, 2024.
- [18] N. Burrell and G. Schoenebeck, "Spot check equivalence: an interpretable metric for information elicitation mechanisms," in *Proc. ACM Web Conf. (WWW)*, 2024, pp. 1175–1186.